

directions. Lemma 3.20 exactly does the job (and, indeed, it was proven in [273] exactly for this purpose). Yet, the transport plans γ_ε we build in the approximation are not optimal for the cost $|x - y|$ but for some costs $|x - y| + \varepsilon|x - y|^2$. We need to do this in order to force the selected limit optimal transport to be the monotone one (through a secondary variational problem, say). Anyway, this will not be an issue: these approximating optimal transport plans will share the same geometric properties that will imply disjointness for the sets $\Omega_i(t)$. In particular, we can prove the following estimate.

Lemma 4.22. *Let γ be an optimal transport plan between μ and an atomic measure ν for a transport cost $c(x, y) = h(y - x)$ where $h : \mathbb{R}^d \rightarrow \mathbb{R}$ is a strictly convex function. Set as usual $\mu_t = (\pi_t)_\# \gamma$. Then we have $\|\mu_t\|_{L^p} \leq (1 - t)^{-d/p'} \|\mu\|_{L^p}$.*

Proof. The result is exactly the same as in Theorem 4.20, where the key tool is the fact that μ_t coincides on every set $\Omega_i(t)$ with a homothety of μ_0 . The only fact that must be checked again is the disjointness of the sets $\Omega_i(t)$.

To do so, take a point $x \in \Omega_i(t) \cap \Omega_j(t)$. Hence there exist x_i, x_j belonging to Ω_i and Ω_j , respectively, so that $x = (1 - t)x_i + ty_i = (1 - t)x_j + ty_j$, being $y_i = T(x_i)$ and $y_j = T(x_j)$ two atoms of ν . But this would mean that $T_t := (1 - t)\text{id} + tT$ is not injective, which is a contradiction to the following Lemma 4.23. Hence the sets $\Omega_i(t)$ are disjoint and this implies the bound on μ_t . $\square$ $\square$

Lemma 4.23. *Let γ be an optimal transport plan between μ and ν for a transport cost $c(x, y) = h(y - x)$ where $h : \mathbb{R}^d \rightarrow \mathbb{R}$ is a strictly convex function, and suppose that it is induced by a transport map T . Choose a representative of T such that $(x, T(x)) \in \text{spt}(\gamma)$ for all x . Then the map $x \mapsto (1 - t)x + tT(x)$ is injective for $t \in]0, 1[$.*

Proof. Suppose that there exist $x \neq x'$ such that

$$(1 - t)x + tT(x) = (1 - t)x' + tT(x') = x_0.$$

Set $a = T(x) - x$ and $b = T(x') - x'$. This also means $x = x_0 - ta$ and $x' = x_0 - tb$. In particular, $x \neq x'$ implies $a \neq b$.

The c -cyclical monotonicity of the support of the optimal γ implies

$$h(a) + h(b) \leq h(T(x') - x) + h(T(x) - x') = h(tb + (1 - t)a) + h(ta + (1 - t)b).$$

Yet, $a \neq b$, and strict convexity imply

$$h(tb + (1 - t)a) + h(ta + (1 - t)b) < th(b) + (1 - t)h(a) + th(a) + (1 - t)h(b) = h(a) + h(b),$$

which is a contradiction. $\square$ $\square$

Theorem 4.24. *Suppose that μ and ν are probability measures on Ω , both belonging to $L^p(\Omega)$, and σ the unique transport density associated to the transport of μ onto ν . Then σ belongs to $L^p(\Omega)$ as well.*

Proof. Let us consider the optimal transport plan $\bar{\gamma}$ from μ to ν defined by (3.4). We know that this transport plan may be approximated by plans γ^ε which are optimal for

the cost $|x - y| + \varepsilon|x - y|^2$ from μ to some discrete atomic measures ν_ε . The corresponding interpolation measures μ_t^ε satisfy the L^p estimate from Lemma 4.22 and, at the limit, we have

$$\|\mu_t\|_{L^p} \leq \liminf_{\varepsilon \rightarrow 0} \|\mu_t^\varepsilon\|_{L^p} \leq (1-t)^{-d/p'} \|\mu\|_{L^p}.$$

The same estimate may be performed from the other direction, since the very same transport plan $\bar{\gamma}$ may be approximated by optimal plans for the cost $|x - y| + \varepsilon|x - y|^2$ from atomic measures to ν . Putting together the two estimates we have

$$\|\mu_t\|_{L^p} \leq \min \left\{ (1-t)^{-d/p'} \|\mu\|_{L^p}, t^{-d/p'} \|\nu\|_{L^p} \right\} \leq 2^{d/p'} \max \{ \|\mu\|_{L^p}, \|\nu\|_{L^p} \}.$$

Integrating these L^p norms we get the bound on $\|\sigma\|_{L^p}$. □ □

Remark 4.25. The same result could have been obtained in a strongly different way, thanks to the displacement convexity of the functional $\mu \mapsto \|\mu\|_{L^p}^p$. This functional is actually convex along geodesics in the space $W_q(\Omega)$ for $q > 1$ (see Proposition 7.29 where, unfortunately, the notation for p and q is reversed). Then we pass to the limit as $q \rightarrow 1$: this gives the result on the interpolating measures corresponding to the optimal plan which is obtained as a limit of the K_q -optimal plans. This plan is, by the way, $\bar{\gamma}$ again. And the integral estimate comes straightforward. Yet, this requires more sophisticated tools than what developed so far.

Example 4.26. We can provide examples (in dimension higher than one) where the summability of σ is no more than that of μ and ν . Take for instance $\mu = f \cdot \mathcal{L}^d$ and $\nu = g \cdot \mathcal{L}^d$ with $f(x_1, x_2, \dots, x_d) = f_0(x_2, \dots, x_d) \mathbb{1}_{[0,1]}(x_1)$ and $g(x_1, x_2, \dots, x_d) = f_0(x_2, \dots, x_d) \mathbb{1}_{[4,5]}(x_1)$. It is not difficult to check that the transport rays in this example go in the direction of the first coordinate vector, and that the optimal σ is absolutely continuous with density given by $\sigma(x) = f_0(x_2, \dots, x_d) s(x_1)$, where $s: \mathbb{R} \rightarrow \mathbb{R}_+$ is characterized by $s' = \mathbb{1}_{[0,1]} - \mathbb{1}_{[4,5]}$, $s(0) = s(5) = 0$ (which comes from the fact that s would be the transport density for the one-dimensional transport between $\mathbb{1}_{[0,1]}$ and $\mathbb{1}_{[4,5]}$). Hence, from $s = 1$ on $[1, 4]$, one sees that the summability of σ is the same of that of f_0 , which is also the same of those of f and g .

4.4 Discussion

4.4.1 Congested transport

As we saw in Section 4.2, Beckmann's problem can admit an easy variant if we prescribe a positive function $k: \Omega \rightarrow \mathbb{R}_+$, where $k(x)$ stands for the local cost at x per unit length of a path passing through x . This models the possibility that the metric is non-homogeneous, due to geographical obstacles given a priori. Yet, it happens in many situations, in particular in urban traffic (as many readers probably note every day) that this metric k is indeed non-homogeneous, but is not given a priori: it depends on the traffic itself. In Beckmann's language, we must look for a vector field $\mathbf{w}$ optimizing a transport cost depending on $\mathbf{w}$ itself!

The easiest model, chosen by Beckmann (see [28] and later [111]) is to consider the same framework as (BP) but supposing that $k(x) = g(|\mathbf{w}(x)|)$ is a function of the modulus of the vector field $\mathbf{w}$. This is quite formal for the moment (for instance it is not meaningful if $\mathbf{w}$ is a measure, but we will see that in this case we can use better spaces than $\mathcal{M}_{\text{div}}^d$). In this case we would like to solve

$$\min \left\{ \int \mathcal{H}(\mathbf{w}(x)) \, dx \quad : \quad \nabla \cdot \mathbf{w} = \mu - \nu \right\}, \quad (4.13)$$

where $\mathcal{H}(z) = H(|z|)$ and $H(t) = g(t)t$. Note that if H is superlinear (if $g(t)$ tends to ∞ as $t \rightarrow \infty$, i.e. if the congestion effect becomes stronger and stronger when the traffic increases) this problem is well posed in the class of vector fields $\mathbf{w} \in L^1$. For instance, if $g(t) = t$, which is the easiest case one can imagine, we must minimize the L^2 norm under divergence constraints:

$$\min \left\{ \int |\mathbf{w}(x)|^2 \, dx \quad : \quad \mathbf{w} \in L^2(\Omega; \mathbb{R}^d), \nabla \cdot \mathbf{w} = \mu - \nu \right\}.$$

This problem is easily solvable since one can see that the optimal $\mathbf{w}$ must be a gradient (we will develop this computation in a more general framework later), and setting $\mathbf{w} = \nabla u$ one gets $\Delta u = \mu - \nu$. This is complemented with Neumann boundary conditions and allows to find u , and then $\mathbf{w}$.

We want now to discuss the meaning and pertinence of this model, keeping into account the following natural questions:

- is it correct that the coefficient k depends on $|\mathbf{w}|$, or should it rather depend on other traffic quantities, such as i_Q ? (note that $\mathbf{w}$ can have cancellations);
- what is the connection with equilibrium issues? in traffic congestion, typically every agent decides alone which path to choose, and the final traffic intensity is rather the output of a collection of individual choices, and not the result of a global optimization made by a single planner;
- is the example $g(t) = t$ a good choice in the modeling ? this implies $g(0) = 0$, i.e. no cost where there is not traffic.

To start our analysis we would like to present first an equilibrium model developed by Wardrop, [295], on a discrete network.

Traffic equilibria on a finite network The main data of the model are a finite oriented connected graph $G = (N, E)$ modeling the network, and some travel times functions $g_e : \theta \in \mathbb{R}_+ \mapsto g_e(\theta)$ giving, for each edge $e \in E$, the travel time on e when the flow on e is θ . The functions g_e are all nonnegative, continuous, nondecreasing and they are meant to capture the congestion effects (which may be different on the different edges, since some roads may be longer or wider and may have different responses to congestion). The last ingredient of the problem is a transport plan on pairs of nodes $(x, y) \in N^2$ interpreted as pairs of origins/destinations. We denote by $(\gamma_{x,y})_{(x,y) \in N^2}$ this transport plan: $\gamma_{x,y}$ represents the “mass” to be sent from x to y . We denote by $C_{x,y}$ the

set of simple paths on the graph G connecting x to y , so that $C := \cup_{(x,y) \in N^2} C_{x,y}$ is the set of all simple paths. A generic path will be denoted by ω and we will use the notation $e \in \omega$ to indicate that the path ω uses the edge e .

The unknown of the problem is the flow configuration. The edge flows are denoted by $i = (i_e)_{e \in E}$ and the path flows are denoted by $q = (q_\omega)_{\omega \in C}$: this means that i_e is the total flow on edge e and q_ω is the mass traveling on ω . Of course the values i_e and q_ω are nonnegative and constrained by the mass conservation conditions:

$$\gamma_{x,y} = \sum_{\omega \in C_{x,y}} q_\omega, \quad \forall (x,y) \in N^2 \quad (4.14)$$

and

$$i_e = \sum_{\omega \in C: e \in \omega} q_\omega, \quad \forall e \in E, \quad (4.15)$$

which means that i is a function of q . Given the edge flows $i = (i_e)_{e \in E}$, the total cost (to be interpreted as a sort of weighted length) of the path $\omega \in C$ is

$$L_i(\omega) = \sum_{e \in \omega} g_e(i_e). \quad (4.16)$$

In [295], Wardrop defined a notion of noncooperative equilibrium that has been very popular since then among engineers working in the field of congested transport and that may be described as follows. Roughly speaking, a Wardrop equilibrium is a flow configuration such that every actually used path should be an optimal path for the cost L_i defined in (4.16). This leads to

Definition 4.27. A Wardrop equilibrium⁵ is a flow configuration $q = (q_\omega)_{\omega \in C}$ satisfying $q_\omega \geq 0$ and the mass conservation constraints (4.14), such that, when we compute the values i_e with (4.15), for every $(x,y) \in N^2$ and every $\omega \in C_{x,y}$ with $q_\omega > 0$ we have

$$L_i(\omega) = \min_{\omega' \in C_{x,y}} L_i(\omega').$$

A few years after the introduction of this notion by Wardrop, Beckmann, McGuire and Winsten [30] realized that Wardrop equilibria can be characterized by the following variational principle:

Theorem 4.28. *The flow configuration $q = (q_\omega)_{\omega \in C}$ is a Wardrop equilibrium if and only if it solves the convex minimization problem*

$$\min \left\{ \sum_{e \in E} H_e(i_e) : q \geq 0 \text{ satisfies (4.14)} \right\} \quad (4.17)$$

where, for each e , we define H_e to be an antiderivative of g_e : $H'_e = g_e$.

⁵Note that this is just an example of Nash equilibrium with a continuum of players, as we will see in Section 7.4.3 and, more particularly, in Box 7.3.

Proof. Assume that $q = (q_\omega)_{\omega \in C}$ (with associated edge flows $(i_e)_{e \in E}$) is optimal for (4.17). Then, for every admissible $\eta = (\eta_\omega)_{\omega \in C}$ with associated (through (4.15)) edge-flows $(j_e)_{e \in E}$, one has

$$\begin{aligned} 0 &\leq \sum_{e \in E} H'_e(i_e)(j_e - i_e) = \sum_{e \in E} g_e(i_e) \sum_{\omega \in C: e \in \omega} (\eta_\omega - q_\omega) \\ &= \sum_{\omega \in C} (\eta_\omega - q_\omega) \sum_{e \in \omega} g_e(i_e), \end{aligned}$$

so that

$$\sum_{\omega \in C} q_\omega L_i(\omega) \leq \sum_{\omega \in C} \eta_\omega L_i(\omega). \quad (4.18)$$

Minimizing the right-hand side thus yields

$$\sum_{(x,y) \in N^2} \sum_{\omega \in C_{x,y}} q_\omega L_i(\omega) = \sum_{(x,y) \in N^2} \gamma_{x,y} \min_{\omega' \in C_{x,y}} L_i(\omega'),$$

which exactly says that q is a Wardrop equilibrium. To prove the converse, it is enough to see that problem (4.17) is convex so that the inequality (4.18) is indeed sufficient, and not only necessary, for a global minimum. $\square$ $\square$

The previous characterization allows to deduce for free existence results, but also uniqueness for i (not for q) as soon as the functions g_e are strictly increasing (so that the H_e are strictly convex).

Remark 4.29. It would be very tempting to deduce from Theorem 4.28 that equilibria are efficient since they are minimizers of (4.17). Yet, the quantity $\sum_{e \in E} H_e(i_e)$ does not represent the natural total social cost measured by the total time lost in commuting which would rather be

$$\sum_{e \in E} i_e g_e(i_e). \quad (4.19)$$

In general, $\theta g_e(\theta) = H_e(\theta)$ and $g_e = H'_e$ are very different conditions. The efficient transport patterns are minimizers of (4.19) and thus are different from equilibria in general. Efficient and equilibria configurations coincide in the special case of power functions where $H_e(\theta) = a_e \theta^p$. Yet, we must note that this case is not realistic since, for $p > 1$, it implies $g_e(0) = 0$. This means that traveling times vanish if there is no traffic, while one should expect a residual non-zero cost even without traffic. Moreover, a famous counter-example due to Braess (see [77]) shows that it may be the case that adding an extra road with very small cost on which the traveling time is always zero leads to an equilibrium where the total commuting time is increased! This illustrates the striking difference between efficiency and equilibrium, a topic which is very well-documented in the finite-dimensional network setting where it is frequently associated to the literature on the so-called *price of anarchy* (see [266]).

Remark 4.30. In the problem presented in this paragraph, the transport plan γ is fixed. This may be interpreted as a *short-term problem*. Instead, we could consider the *long-term* problem where only the distribution of origins μ and of destinations ν are fixed.

In this case, one also obtains as an optimality condition that γ is efficient, in the sense that it minimizes among transport plans in $\Pi(\mu, \nu)$, the total cost

$$\sum \gamma_{x,y} d_i(x,y) \text{ with } d_i(x,y) := \min_{\omega \in C_{x,y}} L_i(\omega).$$

In the long-term problem where one is allowed to change the assignment as well, equilibria still are characterized by a convex minimization problem, i.e. the same above, where one also optimizes over γ .

Optimization and equilibrium in a continuous framework If we want to generalize the previous analysis to a continuous framework, we will formulate the whole path-dependent transport pattern in terms of a probability measure Q on the set of paths (this is the continuous analogue of the path flows $(q_\omega)_\omega$ of the previous paragraph). Then, we will measure the traffic intensity generated by Q at every point x by using the measure i_Q of Section 4.2.3 (this is the continuous analogue of the arc flows $(i_e)_e$ of the previous paragraph). The last and main idea will be to model the congestion effects through a metric that is monotone increasing in the traffic intensity (the analogue of $g_e(i_e)$).

Avoiding technicalities and choosing an informal presentation (see [115] for more details), we take Ω a domain in $\mathbb{R}^d$ (typically, $d = 2$) and

- either two probability measures μ and ν (distribution of sources and destinations) on Ω in the case of the long-term problem,
- or a transport plan γ (joint distribution of sources and destinations, i.e. a joint probability on $\Omega \times \Omega$) in the short-term case,
- or more generally a convex and closed subset $\Gamma \subset \Pi(\mu, \nu)$ and we accept any $\gamma \in \Gamma$ (this is just a common mathematical framework for the two previous cases, where we can take $\Gamma = \{\gamma\}$ or $\Gamma = \Pi(\mu, \nu)$).

We will use the notations of Section 4.2.3, and use probability measures Q on $\mathcal{C} := \text{AC}(\Omega)$, compatible with mass conservation, i.e. such that $(e_0, e_1)_\# Q \in \Gamma$. We shall denote by $\mathcal{Q}(\Gamma)$ the set of admissible Q .

The traffic intensity associated to $Q \in \mathcal{Q}(\Gamma)$ is by definition the measure i_Q defined in Section 4.2.3 and congestion effects are captured by the *metric* associated to Q : suppose $i_Q \ll \mathcal{L}^d$ and set

$$k_Q(x) := g(x, i_Q(x))$$

for a given increasing function $g(x, \cdot) : \mathbb{R}_+ \rightarrow \mathbb{R}_+$. The fact that there exists at least one $Q \in \mathcal{Q}(\Gamma)$ such that $i_Q \ll \mathcal{L}^d$ is not always true and depends on Γ . We will see later how to guarantee it⁶.

In order to define the notion of Wardrop equilibrium in this continuous case, let us define

$$L_{k_Q}(\omega) = \int_0^1 g(\omega(t), i_Q(\omega(t))) |\omega'(t)| dt$$

⁶Note that, even for $\Gamma = \{\gamma\}$ (which is the most restrictive case), assuming $\mu, \nu \in L^\infty$, considerations from incompressible fluid mechanics in [84] allow to build a Q such that $i_Q \in L^\infty$.

and

$$d_{k_Q}(x, y) := \inf_{\omega \in \mathcal{C}, \omega(0)=x, \omega(1)=y} L_{k_Q}(\omega).$$

Paths in $\mathcal{C}$ such that $d_{k_Q}(\omega(0), \omega(1)) = L_{k_Q}(\omega)$ are called geodesics (for the metric induced by the congestion effect generated by Q).

We can define

Definition 4.31. A Wardrop equilibrium is a $Q \in \mathcal{Q}(\Gamma)$ such that

$$Q(\{\omega : L_{k_Q}(\omega) = d_{k_Q}(\omega(0), \omega(1))\}) = 1. \quad (4.20)$$

Existence, and even well-posedness (what does it mean $L_k(\omega)$ if k is only measurable and ω is a Lipschitz curve?) of these equilibria are not straightforward. However, it is possible to characterize equilibria as solutions of a minimal traffic problem.

The full proof is quite involved since it requires to take care of some regularity issues in details. In particular, the use of the weighted length functional $L_{\bar{k}}$ and thus also the geodesic distance $d_{\bar{k}}$ require some attention when $\bar{k}$ is not continuous or at least l.s.c. In [115] a possible construction when $\bar{k}$ is just L^q is given.

Problem 4.32. Let us consider the (convex) variational problem

$$(WP) \quad \min \left\{ \int_{\Omega} H(x, i_Q(x)) dx : Q \in \mathcal{Q}(\Gamma) \right\} \quad (4.21)$$

where $H'(x, \cdot) = g(x, \cdot)$, $H(x, 0) = 0$.

Under some technical assumptions that we do not reproduce here, the main result of [115] is

Theorem 4.33. (WP) admits at least one minimizer. Moreover $\bar{Q} \in \mathcal{Q}(\Gamma)$ solves (WP) if and only if it is a Wardrop equilibrium and $\gamma_Q := (e_0, e_1)_{\#} Q$ solves the optimization problem

$$\min \left\{ \int_{\Omega \times \Omega} d_{k_Q}(x, y) d\gamma(x, y) : \gamma \in \Gamma \right\}.$$

In particular, if Γ is a singleton, this last condition does not play any role (there is only one competitor) and we have existence of a Wardrop equilibrium corresponding to any given transport plan γ . If, on the contrary, $\Gamma = \Pi(\mu, \nu)$, then the second condition means that γ solves a Monge-Kantorovich problem for a distance cost depending on Q itself, which is a new equilibrium condition.

As in the finite-dimensional network case, Wardrop equilibria have a variational characterization which is in principle easier to deal with than the definition. Unfortunately, the convex problem (WP) may be difficult to solve since it involves measures on sets of curves: it means two layers of infinite dimensions!

We finish this section by proposing two recent developments, with the aim of transforming the problem into simpler ones. The first development, easier to present in the short-term case, gives a dual formulation, and provides interesting numerical methods. The second, only in the long-term case, gets rid of the formulation in terms of measures over curves and transforms the problem into a minimal flow problem in the spirit of Beckmann.

Duality and numerics for the short-term problem We want here to give a tractable formulation of the variational problem for the short-term version of (WP). For every $x \in \Omega$ and $k \geq 0$, let us define $H^*(x, \cdot)$ as the Legendre transform of $H(x, \cdot)$. Let us now define the functional

$$J(k) = \int_{\Omega} H^*(x, k(x)) \, dx - \int_{\Omega \times \Omega} d_k(x, y) \, d\gamma(x, y) \quad (4.22)$$

where, again, d_k is the geodesic distance associated to the metric k . Consider

$$\sup \{-J(k) : k \geq 0\}. \quad (4.23)$$

The above maximization problem is important because of the following duality result.

Proposition 4.34.

$$\min(4.21) = \max(4.23) \quad (4.24)$$

and k solves (4.23) if and only if $k = k_Q$ for some $Q \in \mathcal{Q}(\gamma)$ solving (WP).

Moreover, the optimization problem in k is “dimensionally smaller” than that on Q (in the purely continuous case we optimize among scalar functions instead of measures over curves, and if we discretize in space and compute the dimensions we really have a huge difference between the two formulations⁷).

Remark 4.35. Under reasonable assumptions, the dual problem (4.23) has a unique solution. The equilibrium metric k_Q and the equilibrium intensity of traffic i_Q are unique, although Wardrop equilibria Q might not be unique.

Starting from this duality result, in [44] and [45] a consistent numerical scheme to approximate the equilibrium metric k_Q was designed. The idea is to run a descent method on the dual. Hence, one needs to know how to compute (i.e. approximate) the values of d_k , which can be done in an efficient way by the *Fast Marching Algorithm*, but also to differentiate them according to k .

The functional J in (4.22) is considered by means of a discretization grid, and the values of k are considered as defined at the nodes of the grid. The first integral becomes a sum on all the points of the grid, while, for the second, one needs to replace the transport plan γ with a discretized one defined on pairs of points (x, y) on the same grid, and to define $d_k(x, y)$ consequently.

To define such a distance $d_k(x_0, \cdot)$, for a fixed source x_0 , as a function of the second variable, one uses the fact that it is the unique viscosity solution u of the Eikonal non-linear PDE

$$\begin{cases} |\nabla u| = k, \\ u(x_0) = 0, \end{cases} \quad (4.25)$$

Box 4.5. – Good to know! – Discretization of the Eikonal equation and FMM

⁷Note that the same duality trick is also used in the discrete problem over networks, where solving the dual problem is much more convenient than the original one.

To approximate the geodesic distance $u = d_k(x_0, \cdot)$, we discretize both the (unknown) values of u and of k on a square lattice with n points per side. Then we define

$$\begin{aligned} D_1 u^{i,j} &:= \max\{(u^{ij} - u^{i-1,j}), (u^{ij} - u^{i+1,j}), 0\}/h, \\ D_2 u^{i,j} &:= \max\{(u^{ij} - u^{i,j-1}), (u^{ij} - u^{i,j+1}), 0\}/h. \end{aligned}$$

As proposed by Rouy and Tourin [267], the discrete geodesic distance $u = (u^{ij})_{i,j}$ is found as the solution of the following system that discretizes (4.25)

$$Du = k \quad \text{where} \quad Du^{i,j} = \sqrt{(D_1 u^{i,j})^2 + (D_2 u^{i,j})^2}. \quad (4.26)$$

Rouy and Tourin [267] showed that this discrete geodesic distance u converges to u_k when h tends to 0. Then, we have to choose a clever way to solve the above non-linear system. One of the most efficient way is the so-called Fast Marching Algorithm (see [283]): at every step of the algorithm the values $u^{i,j}$ of a bunch of points of the grid have already been computed and the corresponding points are colored in red, and all the other values are set to $+\infty$; then, a tentative computation for all the neighbors of the red point is done, using Equation (4.26) for each of them separately (including the $+\infty$ values of the other points); then, only the smallest among the tentative values is validated, and the corresponding point is colored in red, and the algorithm goes on. This solves the System (4.26) in $O(N \log N)$ steps, where $N = n^d$ is the number of points in the grid.

Once we are able to compute the value of $J(k)$ for every discrete metric k on the grid, we want to differentiate it w.r.t. k , so as to take advantage of a gradient descent algorithm. Actually, one can see that J is not always differentiable in k , but, since all the terms $c_k(x, y)$ may be proven to be concave in k , we face a convex optimization problem, and we can look at the subdifferential of J . Differentiating the equations in (4.26) (see [45]) one gets a new set of equations on the gradient $\nabla_k c_k(x, y)$. The same loop of the Fast Marching Algorithm allows to solve them in a quite efficient way, thus giving an element of the subdifferential⁸

An example of Wardrop equilibria computed with this kind of approach is given in the following Figure 4.4.

Beckmann reformulation of the long-term problem In the long-term problem (4.21), we have one more degree of freedom since the transport plan is not fixed. This will enable us to reformulate the problem as a variational divergence constrained problem *à la* Beckmann and ultimately to reduce the equilibrium problem to solving some nonlinear PDE⁹.

As we already did in Section 4.2.3, for any $Q \in \mathcal{Q}(\Gamma)$ we can take the vector-field $\mathbf{w}_Q$. If we consider the scalar problem (WP), it is easy to see that its value is larger than

⁸This procedure is just a particular case of what is usually called *forward automatic differentiation*.

⁹We also observe that this reduction to a divergence-constrained convex minimization problem allows to provide alternative numerical approaches, as it is done in [39], in the same spirit of the Benamou-Brenier Augmented Lagrangian technique, see also Section 6.1

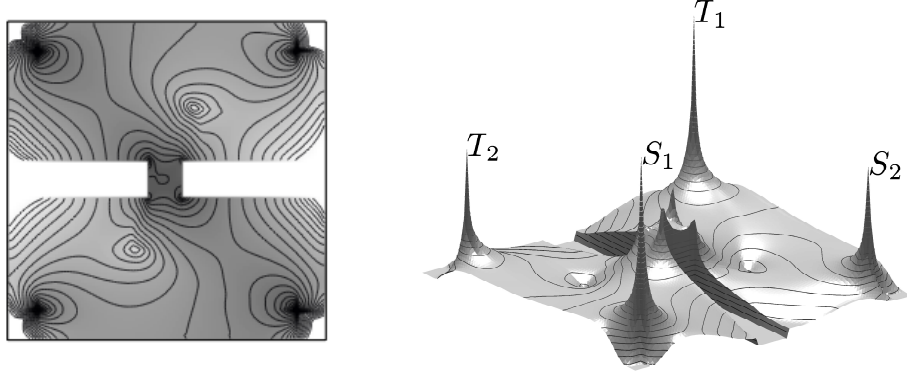


Figure 4.4: Traffic intensity at equilibrium in a city with a river and a bridge, with two sources S_1 and S_2 , and two targets T_1 and T_2 . Picture taken from [45] with permission.

that of the minimal flow problem à la Beckmann:

$$(\text{BP} - \text{cong}) \quad \min \left\{ \int_{\Omega} \mathcal{H}(\mathbf{w}(x)) dx : \nabla \cdot \mathbf{w} = \mu - \nu \right\}, \quad (4.27)$$

where $\mathcal{H}(\mathbf{w}) = H(|\mathbf{w}|)$ and H is taken independent of x only for simplicity. The inequality is justified by two facts: minimizing over all vector fields $\mathbf{w}$ with prescribed divergence gives a smaller result than minimizing over the vector fields of the form $\mathbf{w}_Q$, and then we use $|\mathbf{w}_Q| \leq i_Q$ and the fact that H is increasing.

Actually, the two problems are equivalent. The main ingredient to prove it is Theorem 4.10. If we take a minimizer $\mathbf{w}$ for this minimal flow problem, then we are able to build a measure Q and, as we did in Theorem 4.13, the optimality of $\mathbf{w}$ gives $\mathbf{w} = \mathbf{w}_Q$ and $|\mathbf{w}_Q| = i_Q$, thus proving that the minimal values are the same. Also, we can build a minimizer $\mathbf{w}$ from a minimizer Q (just take $\mathbf{w} = \mathbf{w}_Q$) and conversely a minimizer Q from $\mathbf{w}$ (use Theorem 4.10).

Moreover, we also know from the proof of Theorem 4.10 that the main tool to construct the measure Q is the Dacorogna-Moser flow. If we want to use it to define a unique and canonical flow connecting the two measures μ and ν , then we would like to have some regularity of the vector field $\mathbf{v}$ defined as $\mathbf{v}_t(x) = \mathbf{w}(x)/f_t(x)$, where $f_t = (1-t)\mu + t\nu$. If $\mathbf{v}$ was Lipschitz continuous, then the Cauchy problem

$$\begin{cases} y'_x(t) = \mathbf{v}_t(y_x(t)) \\ y_x(0) = x \end{cases},$$

would have a unique solution for every initial datum x , and would allow to find the measure Q (which would be unique, by the way). Obviously, we can decide to add some assumptions on μ and ν , which will be supposed to be absolutely continuous with regular densities (at least Lipschitz continuous and bounded from below).

However, one needs to prove regularity for the optimal $\mathbf{w}$, and for this one needs to look at the optimality conditions satisfied by $\mathbf{w}$ as a minimizer of (4.27). By considering

perturbations of the form $\mathbf{w} + \varepsilon \tilde{\mathbf{w}}$ with $\nabla \cdot \tilde{\mathbf{w}} = 0$, one gets that the optimal $\mathbf{w}$ must satisfy $\int \nabla \mathcal{H}(\mathbf{w}) \cdot \tilde{\mathbf{w}} = 0$ for any divergence-free vector field $\tilde{\mathbf{w}}$. This means that $\nabla \mathcal{H}(\mathbf{w})$ is a gradient (see Box 6.2 in Section 6.2, which has to be adapted to an L^p setting¹⁰). We can call it ∇u , and, by inverting $\nabla \mathcal{H}$, we get that the minimizers of (4.27) satisfy $\mathbf{w} = \nabla \mathcal{H}^*(\nabla u)$ where $\mathcal{H}^*$ is the Legendre transform of $\mathcal{H}$ and u solves the PDE:

$$\begin{cases} \nabla \cdot (\nabla \mathcal{H}^*(\nabla u)) &= \mu - \nu, & \text{in } \Omega, \\ \nabla \mathcal{H}^*(\nabla u) \cdot \mathbf{n} &= 0, & \text{on } \partial\Omega. \end{cases} \quad (4.28)$$

This equation turns out to be a standard Laplace equation if $\mathcal{H}$ is quadratic, or it becomes a p -Laplace equation for other power functions. In these cases, regularity results are well-known, under regularity assumptions on μ and ν .

Remark 4.36. Note that the above considerations allow to answer to the question “Given μ, ν , when does a traffic plan Q with $i_Q \in L^p$ exists?”. The answer is yes if and only if $\mu - \nu \in (W^{1,q})^*$, and the proof is left as an exercise, see **Ex**(26).

As we already mentioned, the case of power-like functions H is not suitable for modeling reasons. Recall that $H' = g$ where g is the congestion function, so it is natural to have $g(0) > 0$: the metric should be positive even if there is no traffic! This means that the radial function $\mathcal{H}$ cannot be differentiable at 0 and then its subdifferential at 0 contains a ball. By duality, this implies $\nabla \mathcal{H}^* = 0$ on this ball which makes (4.28) very degenerate, even worse than the p -Laplacian. For instance, a reasonable model of congestion is $g(t) = 1 + t^{p-1}$ for $t \geq 0$, with $p > 1$, so that

$$\mathcal{H}(\mathbf{w}) = \frac{1}{p} |\mathbf{w}|^p + |\mathbf{w}|, \quad \mathcal{H}^*(z) = \frac{1}{q} (|z| - 1)_+^q, \quad \text{with } q = p' = \frac{p}{p-1} \quad (4.29)$$

so that the optimal $\mathbf{w}$ is

$$\mathbf{w} = \left(|\nabla u| - 1 \right)_+^{q-1} \frac{\nabla u}{|\nabla u|},$$

where u solves the very degenerate PDE:

$$\nabla \cdot \left(\left(|\nabla u| - 1 \right)_+^{q-1} \frac{\nabla u}{|\nabla u|} \right) = \mu - \nu, \quad (4.30)$$

with Neumann boundary condition

$$\left(|\nabla u| - 1 \right)_+^{q-1} \frac{\nabla u}{|\nabla u|} \cdot \mathbf{n} = 0.$$

Note that there is no uniqueness for u but there is for $\mathbf{w}$.

For this degenerate equation we cannot expect regularity results which go beyond Lipschitz continuity for u (indeed, since the diffusion coefficient identically vanishes in the zone where $|\nabla u| \leq 1$, every $u \in \text{Lip}_1$ is solution of the corresponding homogeneous equation). On the other hand, we can expect better regularity for $\mathbf{w}$ (roughly speaking, $\mathbf{w}$ is obtained by gluing two zones: the one where $\mathbf{w} = 0$ and the one where the equation

¹⁰Observe that the same result can also be directly obtained from duality arguments, as it is done in Theorem 2.1 in [82]

is elliptic). However, we will not be able to obtain $\mathbf{w}$ itself to be Lipschitz. On the other hand, Sobolev regularity of $\mathbf{w}$ and Lipschitz regularity results for u are proven in [82]. This enables one to build a flow *à la* DiPerna-Lions [157] and then to justify rigorously the existence of a unique and well-defined flow for a.e. initial datum x . Interestingly, recent continuity results are also available (see [279] in dimension 2, and then [127], with a different technique in arbitrary dimension), obtained as a consequence of a fine analysis of this degenerate elliptic PDE. Besides the interest for this regularity result in itself, we also stress that continuity for $\mathbf{w}$ implies continuity for the optimal i_Q , and this exactly gives the regularity which is required in the proof of Theorem 4.33 (the main difficulty being defining $c_{\bar{k}}$ for a non-continuous $\bar{k}$, and this is the reason why our previous arguments are only formal).

4.4.2 Branched transport

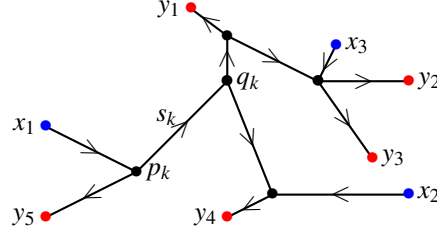
The goal of this second part of the discussion section is to present an opposite minimal flow problem, where, contrary to what happens in traffic congestion, joint transportation on a common path is indeed encouraged. This is reasonable when we consider “economy of scale” principles, and it is also something that we can observe in many different phenomena. In many cases, when a single planner needs to organize a transport (building a transportation network, for instance), the total cost happens to be *sub-additive* (the cost for moving two objects cannot exceed the sum of the two costs), and in many cases it has *decreasing marginal costs* (i.e. the cost for adding a unit to a given background quantity is a decreasing function of the background, which means that the cost is actually concave).

Note that we are considering here the transport cost as a function of the mass, and not of the length. The silent assumption is that the cost will be linear in the length (as it happens in all minimal flow problems of this chapter). But it will be concave in the mass, opposite to what we saw in Section 3.3.2.

Many branching structures transporting different fluids, such as road systems, communication networks, river basins, blood vessels, leaves, trees and so on... may be easily thought of as coming from a variational principle for a cost of this kind.

Note that, exactly as it happened for traffic congestion, modeling this kind of effects requires, either in Lagrangian or Eulerian language, to look at the paths actually followed by each particle. It could not be done with the only use of a transport plan $\gamma \in \Pi(\mu, \nu)$. Yet, once we choose the good formulation via the tools developed in this chapter, we can guess the shape of the optimal solution for this kind of problem: particles are collected at some points, move together as much as possible, and then branch towards their different destinations. This is why this class of problems is nowadays known as “branched transport”.

Recently these problems received a lot of attention by mathematicians, but in fact the formalization of the discrete framework, which can be expressed in terms of Dirac masses, is very classical in optimization and operational research. As we did for congested transport, we start from the discrete framework, which was introduced by Gilbert, [186]. We then move on to the continuous models, which are much more recent and were partly developed and studied by Bernot, Caselles, Morel [50, 51] and



Maddalena, Solimini [221, 222] for the *Lagrangian* model, and by Xia [297], who gave an *Eulerian* continuous extension of the Gilbert Energy.

The discrete model Translating into our language the work by Gilbert [186], we consider two atomic probability measures $\mu = \sum_{i=1}^m a_i \delta_{x_i}$ and $\nu = \sum_{j=1}^n b_j \delta_{y_j}$ supported in some compact subset $\Omega \subset \mathbb{R}^d$, and we want to connect them through a weighted oriented graph G . A *weighted oriented graph* in $\mathbb{R}^d$ is just a system $G = (s_k, \theta_k)_k$ where s_k is a pair of points $s_k = (p_k, q_k)$ representing an oriented segment in $\mathbb{R}^d$ and $\theta_k \in \mathbb{R}_+$ is a weight associated to the segment s_k . We denote by $\hat{s}_k$ the orientation $\frac{q_k - p_k}{|q_k - p_k|}$ of s_k (provided it is not trivial) and $|s_k|$ its length $|q_k - p_k|$. The points p_k, q_k are called the *vertices* of the graph and the segments s_k are its *edges*.

Given μ, ν , we say that G irrigates ν from μ if it satisfies the well-known Kirchhoff Law for electric circuits: for each vertex x of the graph, one must have

$$\text{incoming mass at } x = \text{outcoming mass at } x.$$

We count as *incoming* the mass conveyed by the segments $s_k = (p_k, q_k)$ reaching x , i.e. such that $x = q_k$, increased by the mass given by μ , i.e. increased by a_i if $x = x_i$. Similarly, the *outcoming mass* at x is the total weight of the segments $s_k = (p_k, q_k)$ leaving x , i.e. such that $p_k = x$, decreased by the mass taken out by ν , i.e. by b_j if $x = y_j$. The set of graphs irrigating ν from μ is denoted by $G(\mu, \nu)$.

For $\alpha \in [0, 1]$, we define the cost of α -irrigation of a graph G , also called *Gilbert energy* as

$$\mathcal{E}_\alpha(G) = \sum_{k=1}^n \theta_k^\alpha |s_k|,$$

which means that the cost of moving a mass m along a segment of length l is $m^\alpha l$.

Given μ, ν atomic probabilities, we want to minimize this energy among all graphs sending μ to ν , which reads

$$\min \{ \mathcal{E}_\alpha(G) : G \in G(\mu, \nu) \}.$$

This problem was introduced by Gilbert in [186, 187] as an extension of Steiner's minimal length problem which corresponds¹¹ to the case $\alpha = 0$, where one wants to

¹¹To have an exact equivalence between the Steiner minimal connection problem and Gilbert problem with $\alpha = 0$, one needs to consider a measure μ composed of a unique Dirac mass, so that every point in $\text{spt}(\nu)$ must be connected to it, hence getting connectedness of the graph.

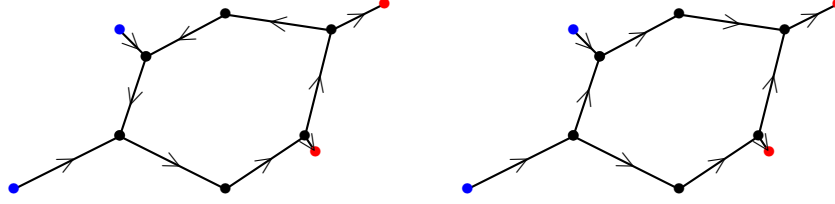


Figure 4.5: An example of a strong cycle (left) and of a cycle which is not strong (right)

minimize the total length of the network. The main applications that Gilbert referred to were in the field of communication networks.

Note that for $\alpha \in [0, 1]$ the function $t \mapsto t^\alpha$ is concave and subadditive; it is strictly concave for $\alpha \in]0, 1[$. In this way larger tubes bringing the mass from μ to ν are preferred to several smaller ones transporting the same total mass.

It is not evident to see why this problem admits a minimizer. Indeed, we minimize among finite objects, but we do not impose uniform bounds (on the number of edges, of vertices...). We need to give an important definition¹²

Definition 4.37. We call *cycle* (resp. *strong cycle*) of G any sequence of segments (resp. oriented segments) which are adjacent.

As the function $t \mapsto t^\alpha$ is increasing, we can easily reduce the energy of a graph by removing strong cycles. It is not difficult to check that the concavity of the same function also ensures that one reduces the energy by removing any cycle. Therefore one can take a minimizing sequence of graphs which have no cycles, i.e. which are trees, and this implies a uniform bound on the number of edges (in terms of the total number of atoms of μ and ν). This provides compactness and proves the existence of a minimizer. Any minimizer will have no strong cycles, and no cycles if $\alpha \in]0, 1[$.

Moreover, it is possible to prove an optimality condition on the free junctions in the graph. This means that, at every node x of the graph which is not one of the points x_i or y_j , a balance of the direction must hold: the condition is

$$\sum_{k: x=p_k} \theta_k^\alpha \hat{s}_k - \sum_{k: x=q_k} \theta_k^\alpha \hat{s}_k = 0, \quad (4.31)$$

which means that the sum of the unit vectors corresponding to all the segments touching the point x , always taken as exiting from x independently of the true orientation $\hat{s}_k$, each weighted with θ_k^α , must vanish¹³.

A Relaxed Continuous Model (Xia) More recently Xia, in [297], has proposed a new formalization leading to generalizations of this problem to arbitrary probability

¹²Unfortunately, to cope with the language usually adopted in branched transport, we cannot be completely coherent with the rest of the chapter, where we called “cycles” what we call here “strong cycles”.

¹³For triple junctions, which are the most common ones, this gives interesting results: when $\alpha = 0$ we get the well-known condition about Steiner trees, with three 120° angles, and for $\alpha = 0.5$ we have a 90° angle (see Ex(30)); yet, in general, the angles depend on the masses θ_k

measures μ and ν . The main idea was to translate the Kirchhoff law constraint into a divergence constraint. To be precise, and since we will need this language later on, we introduce some heavy notation.

If $E \subset \Omega$ is an 1-rectifiable set¹⁴, $\tau : E \rightarrow \mathbb{S}^{d-1}$ is a measurable vector field, and $\theta : E \rightarrow \mathbb{R}_+$ is $\mathcal{H}^1$ -integrable, we define the finite vector measure $[E, \tau, \theta] \in \mathcal{M}^d(\Omega)$ through

$$\langle [E, \tau, \theta], \xi \rangle = \int_E \theta(x) \xi(x) \cdot \tau(x) d\mathcal{H}^1(x),$$

for all $\xi \in C(\Omega; \mathbb{R}^d)$.

Given a finite graph $G = (s_k, \theta_k)_k$, we associate to G the measure

$$\mathbf{w}_G := \sum_k [s_k, \hat{s}_k, \theta_k] \in \mathcal{M}^d(\Omega).$$

We can express both the cost and the constraint in terms of $\mathbf{w}_G$ only. First note that $G \in \mathbf{G}(\mu, \nu)$ (i.e. it satisfies mass balance conditions for irrigating ν from μ) if and only if $\nabla \cdot \mathbf{w}_G = \mu - \nu$. Then, we also write

$$\mathcal{E}_\alpha(G) = \sum_{k=1}^n \theta_k^\alpha |s_k| = \int_K |\theta(x)|^\alpha d\mathcal{H}^1(x),$$

where we called θ the density of $\mathbf{w}_G$ w.r.t. the $\mathcal{H}^1$ measure.

This suggests a continuous Eulerian formulation of Gilbert's problem by generalizing Gilbert's energy, the minimization problem becoming

$$(XP) \quad \min \{M_\alpha(\mathbf{w}) : \mathbf{w} \in \mathcal{M}_{\text{div}}^d(\Omega), \nabla \cdot \mathbf{w} = \mu - \nu\}. \quad (4.32)$$

Obviously, one needs to define the energy M_α (which is also called the “ α -mass” of $\mathbf{w}$). The original idea by Xia is to define it by relaxation, i.e.

$$M_\alpha(\mathbf{w}) := \inf \left\{ \liminf_n \mathcal{E}_\alpha(G_n) : G_n \in \mathbf{G}(\mu_n, \nu_n), \mathbf{w}_{G_n} \rightharpoonup \mathbf{w} \text{ in } \mathcal{M}_{\text{div}}^d(\Omega) \right\},$$

where the convergence in $\mathcal{M}_{\text{div}}^d(\Omega)$ means $\mathbf{w}_{G_n} \rightharpoonup \mathbf{w}$ and $\nabla \cdot \mathbf{w}_{G_n} = \mu_n - \nu_n \rightharpoonup \nabla \cdot \mathbf{w}$ as measures.

As a consequence of a famous rectifiability theorem by B. White, [296], one can prove that $M_\alpha(\mathbf{w}) < +\infty$ implies that $\mathbf{w}$ is 1-rectifiable¹⁵ and, with a finer analysis, we get

$$M_\alpha(\mathbf{w}) = \begin{cases} \int_E |\theta(x)|^\alpha d\mathcal{H}^1(x) & \text{if } \mathbf{w} = [E, \tau, \theta] \\ +\infty & \text{otherwise.} \end{cases}$$

¹⁴We already met rectifiable sets when studying the differentiability of convex functions, in Chapter 1: 1-rectifiable sets are defined as those sets which are covered, $\mathcal{H}^1$ -a.e., by a countable union of Lipschitz curves. Anyway, the reader can easily pretend that “1-rectifiable” is a synonym of “1-dimensional” and it should be enough to follow the rest of the discussion.

¹⁵We define rectifiable vector measures as those which can be expressed in the form $[E, \tau, \theta]$ with E rectifiable. This language is borrowed from that of rectifiable currents, but currents can be considered vector measures. The reader can look at [162] for more details.

It should be proven that, when μ and ν are both atomic measures, we retrieve the problem by Gilbert (this is not trivial, as we admitted lots of new competitors).

Compared to the finitely atomic case, another difference is the finiteness of the minimum in (XP). Indeed, it is always possible to connect two atomic measures with a finite graph, and the cost of finite graphs is always finite. On the contrary, it is not clear for general measures μ, ν that there exists a $\mathbf{w}$ with $M_\alpha(\mathbf{w}) < \infty$ and $\nabla \cdot \mathbf{w} = \mu - \nu$.

Irrigability and Irrigation Distance For $\mu, \nu \in \mathcal{P}(\Omega)$, we set $d_\alpha(\mu, \nu) := \min(\text{XP}) \in [0, +\infty]$ (which obviously depends on μ, ν). First of all we consider the question whether this value is finite. Supposing $0 \in \Omega$, it is clear that, if $d_\alpha(\mu, \delta_0), d_\alpha(\nu, \delta_0) < +\infty$, then $d_\alpha(\mu, \nu) < +\infty$. Hence¹⁶ we will concentrate on the question whether $d_\alpha(\mu, \delta_0) < +\infty$.

One of the main results of [297] is the following.

Proposition 4.38. *If $\alpha > 1 - 1/d$ and $\Omega = [-a, a]^d$, then $d_\alpha(\mu, \delta_0) < +\infty$ for every $\mu \in \mathcal{P}(\Omega)$. This implies $d_\alpha(\mu, \nu) < +\infty$ for every pair (μ, ν) . The same is true in every compact domain, just by fitting it into a large cube.*

Moreover, keeping the conditions $\alpha > 1 - 1/d$ and Ω compact, d_α is actually a distance over $\mathcal{P}(\Omega)$ and it metrizes the weak convergence of probabilities.

We just prove the very first part of this statement.

Proof. The idea to prove $d_\alpha(\mu, \delta_0) < +\infty$ is the following: take a dyadic approximation of μ by partitioning the cube Ω into 2^{nd} small cubes of edge 2^{-n} and putting a Dirac mass of the corresponding measure at the center of each small cube, thus getting an atomic measure μ_n . Then connect μ_n to μ_{n+1} through a finite graph G_n (to which we associate a flow $\mathbf{w}_n$), which sends every Dirac mass of μ_n , corresponding to some small cube D , to the 2^d Dirac masses of μ_{n+1} corresponding to the sub-cubes of D , along straight lines. A simple calculation gives the estimate:

$$M_\alpha(G_n) \leq C \sum_{k=1}^{2^{nd}} a m_k^\alpha 2^{-n},$$

where the masses m_k are the masses contained in each of the cubes at the n -th subdivision. Then we use $\sum_{k=1}^N m_k^\alpha \leq N^{1-\alpha} (\sum_{k=1}^N m_k)^\alpha$ (which comes from Jensen's inequality), and get $\mathcal{E}_\alpha(G_n) \leq aC2^{-n(1-d(1-\alpha))}$.

It is clear by construction that we only need $\sum_n \mathcal{E}_\alpha(G_n) < +\infty$, and this is true as soon as $1 - d(1 - \alpha) > 0$. $\square$

Some counter-examples, that we do not develop here (but one can see **Ex(29)**) complement the previous result and show that the above statement is sharp in terms of α . More generally, we can think that the irrigability of a measure starting from δ_0 is a matter of its dimension, but suitable notions of dimensions (which are not simply the Hausdorff dimension of the support) have to be chosen. These matters are discussed in [154, 155].

¹⁶Indeed, even if not completely evident, it can be proven that, at least in the case where $\text{spt}(\mu) \cap \text{spt}(\nu) = \emptyset$, the condition $d_\alpha(\mu, \nu) < +\infty$ is actually equivalent to $d_\alpha(\mu, \delta_0), d_\alpha(\nu, \delta_0) < +\infty$.

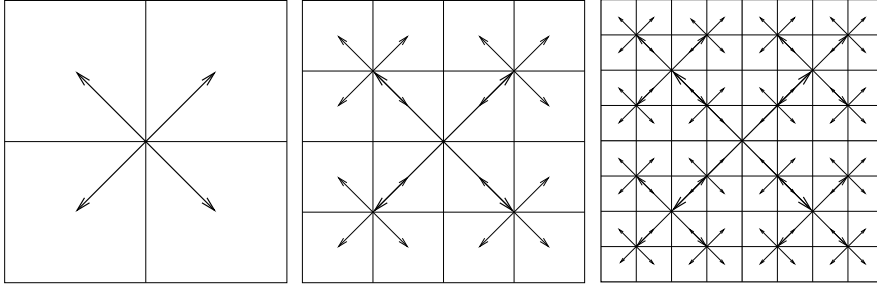


Figure 4.6: Different steps of the construction of Proposition 4.38

It is interesting to note that the above distance d_α may also be obtained by relaxation (see [221]) from the functional which associates to every pair of atomic probability measures $\mu = \sum_i a_i \delta_{x_i}$, $\nu = \sum_j b_j \delta_{y_j}$ the quantity

$$\overline{d}_\alpha(\mu, \nu) := \min \left\{ \sum_{ij} |x_i - y_j| \gamma(\{(x_i, y_j)\})^\alpha : \gamma \in \Pi(\mu, \nu) \right\}.$$

This quantity, which corresponds to a sort of non-linear version of the Kantorovich problem, is only well-defined on atomic measures. Then, we can check that we have

$$d_\alpha = \sup \left\{ d : \mathcal{P}(\Omega) \times \mathcal{P}(\Omega) \rightarrow [0, +\infty] : \begin{cases} d \leq \overline{d}_\alpha \text{ on atomic measures,} \\ d \text{ is l.s.c. for } \rightharpoonup, \\ d \text{ satisfies the triangle inequality.} \end{cases} \right\}. \quad (4.33)$$

The Lagrangian Formulation We want now to provide an alternative formulation of the branched transport problem, based on the use of measures on paths. This formulation¹⁷ has been introduced in [222] and then deeply studied in [50].

As usual, we will be given two measures $\mu, \nu \in \mathcal{P}(\Omega)$ and look for a *traffic plan*¹⁸, i.e. a measure $Q \in \mathcal{P}(\mathcal{C})$, where $\mathcal{C} := \text{AC}(\Omega)$, with $\int_{\mathcal{C}} L(\omega) dQ(\omega) < +\infty$. As usual, we will restrict our attention to those traffic plans satisfying $(e_0)_\# Q = \mu, (e_1)_\# Q = \nu$ (the case where we fix also the coupling $(e_0, e_1)_\# Q = \gamma$ will not be considered, and, by the way, the connection with the previous Eulerian model by Xia does not work if γ is fixed).

To introduce the quantity that we want to minimize, we need to define

$$\theta_Q(x) := Q(\{\omega \in \mathcal{C} : x \in \omega([0, 1])\}), \quad m_Q(x) := \int_{\mathcal{C}} \#(\{t \in [0, 1] : \omega(t) = x\}) dQ(\omega).$$

¹⁷What we provide here is just a translation into the language of this chapter of the model proposed in [222, 50], which uses “parametrized traffic plans” instead of measures on paths, but it is just a matter of language.

¹⁸The terminology has been introduced for this very purpose by the authors of [50].

Proposition 4.39. *The following properties are satisfied.*

1. *We always have $0 \leq \theta_Q(x) \leq m_Q(x) \leq 1$, with equality $\theta_Q = m_Q$ whenever Q is concentrated on loop-free curves.*
2. *The function $(x, Q) \mapsto \theta_Q(x)$ is u.s.c. w.r.t. the convergence of points $x \in \Omega$ and the weak convergence of measures $Q \in \mathcal{P}(\mathcal{C})$.*
3. *The traffic intensity i_Q is equal to $m_Q \cdot \mathcal{H}^1$ whenever there exists a set Γ , σ -finite for the $\mathcal{H}^1$ measure, with $\omega(t) \in \Gamma$ for Q -a.e. curve ω and a.e. time t .*

Then, we define the energy I_α through

$$I_\alpha(Q) := \int_{\mathcal{C}} Z_Q(\omega) dQ(\omega),$$

where

$$Z_Q(\omega) := L_{\theta_Q^{\alpha-1}}(\omega) = \int_0^1 \theta_Q(\omega(t))^{\alpha-1} |\omega'(t)| dt.$$

The problem that we consider is then

$$\min\{I_\alpha(Q) : Q \in \mathcal{P}(\mathcal{C}), (e_0)_\# Q = \mu, (e_1)_\# Q = \nu\}.$$

Note that the trivial estimate $I_\alpha(Q) \geq \int_{\mathcal{C}} L(\omega) dQ(\omega)$ provides compactness on a minimizing sequence (indeed, the energy is invariant under reparametrization of each curve, and hence length bounds turn into Lipschitz bounds, as we did in the rest of the chapter). The existence of a minimizer is then obtained by proving semi-continuity. Indeed, combining fact 2. with a simple Fatou Lemma, we obtain that the function $(Q, \omega) \mapsto Z_Q(\omega)$ is l.s.c. (for the weak convergence on Q and the uniform convergence on ω). Finally we use the following.

Proposition 4.40. *If $f : \mathcal{P}(\mathcal{C}) \times \mathcal{C} \rightarrow \mathbb{R}$ is l.s.c., then $F : Q \mapsto \int_{\mathcal{C}} f(Q, \omega) dQ(\omega)$ is l.s.c. on $\mathcal{P}(\mathcal{C})$.*

Then, we can prove the following facts

Proposition 4.41. 1. *For every Q with $I_\alpha(Q) < +\infty$ the set $\Gamma = \{x : \theta_Q(x) > 0\}$ is σ -finite for the measure $\mathcal{H}^1$ and we have $I_\alpha(Q) = \int_\Gamma \theta_Q^{\alpha-1} m_Q d\mathcal{H}^1$.*

2. *If Q is optimal, then it is concentrated on loop-free curves and $I_\alpha(Q) = \int_\Gamma \theta_Q^\alpha d\mathcal{H}^1$.*

3. *If Q is optimal, then it is cycle-free¹⁹.*

These property, and in the particular the expression of I_α as the integral of the α -power of the multiplicity θ_Q , suggest the connection with the Eulerian model. A precise equivalence can be proven, either via the tools in [50], or via Theorem 4.10.

¹⁹We can give a precise definition in the following way: for $x, y \in \Omega$, define $[x, y] = \{\omega \in \mathcal{C} : \exists s < t \text{ such that } \omega(s) = x \text{ and } \omega(t) = y\}$; we say that Q is cycle-free if there are not $x_1, \dots, x_n$ with $x_1 = x_n$ such that $Q([x_i, x_{i+1}]) > 0$ or $Q([x_{i+1}, x_i]) > 0$ for all $i = 1, \dots, n-1$.

The Single-Source Case It is interesting to see the special features of the problem when we fix the source measure as a single Dirac mass δ_0 . This particular case is already very rich. In this case, it is possible to study the problem with a different definition of multiplicity: indeed, we can expect that whenever two curves pass at a same point x they had traveled together on all the path from 0 to x . In this case, it is better to change a little bit the framework, because parametrizing all the curves on a fixed finite-length integral $[0, 1]$ could be too restrictive. Also, this gives the occasion to observe the original framework presented in [222] (which can be checked to be equivalent to the general one).

We consider now the space $\Lambda = \{\omega \in \text{Lip}_1(\mathbb{R}_+; \Omega), \omega(0) = 0\}$, endowed with the topology of uniform convergence on compact sets (recall that this topology is metrizable by some distance d , and that this makes Λ a compact metric space thanks to the Ascoli Arzelà Theorem).

If $\omega \in \Lambda$, we define its *stopping time* by

$$\text{Stop}(\omega) = \inf\{t \geq 0 : \omega \text{ is constant on } [t, +\infty[\}.$$

We denote by Λ_{stop} the set of curves ω such that $\text{Stop}(\omega) < \infty$. Then we define the following subsets of Λ_{stop} : $\Lambda_{\text{arc}}, \Lambda_{\text{inj}}, \Lambda_{\text{lf}}$, respectively the set of curves which are parameterized by arc length on $[0, \text{Stop}(\omega)]$, which are injective on $[0, \text{Stop}(\omega)[$ and which are injective on $[0, \text{Stop}(\omega)]$ (these curves are called *loop-free*).

We call *traffic pattern* any $\eta \in \mathcal{P}(\Lambda)$ (we use the notation η to distinguish from the case $Q \in \mathcal{P}(\mathcal{C})$) satisfying

$$\int_{\Lambda} \text{Stop}(\omega) d\eta(\omega) < +\infty.$$

We define the multiplicity and the energy according to the following variant of the previous definitions: for the multiplicity, we first define

$$[\omega]_t := \{\tilde{\omega} \in \Lambda_{\text{stop}} : \omega(s) = \tilde{\omega}(s) \quad \forall s \in [0, t]\},$$

the set of curves which coincide with ω up to time t , and for notational simplicity we set $|\omega|_{t, \eta} := \eta([\omega]_t)$. This last quantity will play the role of the multiplicity θ . Then we define

$$\tilde{Z}_{\eta}(\omega) := \int_0^{\text{Stop}(\omega)} |\omega|_{t, \eta}^{\alpha-1} dt$$

and we call η -good any curve ω with $\tilde{Z}_{\eta}(\omega) < +\infty$. Then, we use the energy

$$\tilde{I}_{\alpha}(\eta) := \int_{\Lambda} \tilde{Z}_{\eta}(\omega) d\eta(\omega).$$

The problem becomes the minimization of $\tilde{I}_{\alpha}$ among probabilities $\eta \in \mathcal{P}(\Lambda)$ satisfying $(e_{\infty})_{\#} \eta = \nu$, where $e_{\infty}(\omega) = \omega(\text{Stop}(\omega))$. Note that e_{∞} is well-defined on Λ_{stop} , but any finite energy traffic plan is necessarily concentrated on Λ_{stop} , hence the condition makes sense; moreover, $\eta \mapsto (e_{\infty})_{\#} \eta$ is continuous for the weak convergence when restricted to any set $\{\eta \in \mathcal{P}(\Lambda) : \int_{\Lambda} \text{Stop}(\omega) d\eta(\omega) \leq C\}$.

It is not difficult to prove the existence of a minimizer in this case, and it is also possible to prove that minimizers are concentrated on $\Lambda_{\text{arc}} \cap \Lambda_{\text{inj}}$ and are even loop-free (i.e. we can prove $\eta(\Lambda_{\text{lf}}) = 1$, even if it is less evident). Note that we withdrew from the definition of the energy the term $|\omega'|$ and this forces the minimizers to saturate the constraint $|\omega'| \leq 1$.

Remark 4.42. Other intermediate models may be introduced, all differing in the definition of the multiplicity of the curve ω at time t . See for instance [49] or [221].

The class of η -good curves is an explicitly defined set of curves which shares many of the properties satisfied η -almost everywhere. For instance, if η is optimal and Q is a measure on $\mathcal{C}$ representing η after a suitable reparametrization in time, we have $|\omega|_{t,\eta} = \theta_Q(\omega(t)) = m_Q(\omega(t))$, and this is true for every ω which is η -good.

An important point is the following: if ω_1, ω_2 are η -good and η is optimal, if $(e_\infty)(\omega_1) = (e_\infty)(\omega_2)$, then $\tilde{Z}_\eta(\omega_1) = \tilde{Z}_\eta(\omega_2)$ (see [272]). As a corollary, we get for instance that any η -good curve ω is in fact loop-free, i.e. it is injective on $[0, \text{Stop}(\gamma)]$.

Definition 4.43 (Landscape function). We define the landscape function z_η associated to the traffic pattern η through

$$z_\eta(x) = \begin{cases} Z_\eta(\omega) & \text{if } \omega \text{ is an } \eta\text{-good curve and } x = e_\infty(\omega), \\ +\infty & \text{if there is no } \eta\text{-good curve ending at } x. \end{cases}$$

The introduction of the landscape function ([272]) is justified by several different considerations. On the one hand it plays the role of the first variation of the functional

$$X_\alpha(v) = d_\alpha(\delta_0, v),$$

as we can prove (by concavity)

$$X_\alpha(\mu) \leq X_\alpha(v) + \alpha \int_\Omega z_\eta d(\mu - v),$$

where η is any optimal traffic plan irrigating v . Considering that we also have $X_\alpha(v) = \int z_\eta dv$, it seems to play the same role as the Kantorovich potential plays for the classical Monge transport cost²⁰.

On the other hand, and this justifies its name, z_η is a natural object studied by geophysicists when dealing with river basins (see [263] for details and [272] for a brief mathematical introduction to this theory): in their model, after a first evolution phase, it is assumed that landscape erosion creates a stable configuration where the elevation $z(x)$ at each point is characterized by two properties: the river flows in the direction of maximal slope, and the slope $|\nabla z|$ is proportional to the discharge (i.e. multiplicity) coefficient θ , raised to a power close to $-1/2$ (say $\alpha - 1$, with α close to the two-dimensional critical exponent $1/2$).

It is possible to prove few properties of the landscape function.

²⁰But it does not seem to come from a dual problem.

Theorem 4.44. *The landscape function z is lower semi-continuous. It is Hölder continuous²¹ of exponent $\beta = d(\alpha - (1 - \frac{1}{d}))$ whenever $\alpha > 1 - \frac{1}{d}$, Ω has Lipschitz boundary and $\nu \geq c\mathcal{L}_\Omega$. Moreover, z has maximal slope in the direction of the irrigation network in the following sense: if ω is an η -good curve, $t_0 \leq \text{Stop}(\omega)$, $x_0 = \omega(t_0)$ and $\theta_0 = |\omega|_{t_0, \eta}$, then for any $x \in \Omega$ we have*

$$z(x) \geq z(x_0) - \theta_0^{\alpha-1} |x - x_0| - o(|x - x_0|).$$

Regularity The regularity of the optimal networks in branched transport is a delicate issue, which has been studied both in Eulerian and Lagrangian framework. One of the first results concern what Xia called “interior regularity”: here interior means “far from the supports of μ and ν ”, which can be considered as the boundary of the transport network. These results, proven in [298] with an Eulerian approach and in [51] with the Lagrangian approach, state that, far from these supports, optimal networks are locally finite graphs. In particular, they satisfy the angle law (4.31) at every branching point. The question now becomes whether this angle law is satisfied at an infinitesimal level (in the sense that the directions to be considered are tangent lines and not segments) inside the support of μ and ν . [300] gives partial results in this direction (characterization of the limits up to subsequences), but a full solution is not yet available. On the other hand, in the simpler case of a single source, [241] provides the following result:

Theorem 4.45. *Let ω be an η -good curve, parametrized with constant speed, of an optimal traffic pattern irrigating a measure ν with $a\mathcal{L}_\Omega \leq \nu \leq b\mathcal{L}_\Omega$, for $0 < a < b < \infty$. Suppose that Ω is a Lipschitz bounded domain. Consider $t_0 < \text{Stop}(\omega)$. Let θ_i , $i \in I$ the masses of all trees branching from ω in the interval $[0, t_0]$. Then $\sum_i \theta_i^\alpha < \infty$ and $\omega' \in BV([0, t_0])$. In particular, ω has two half-tangents at all points and a tangent at all points which are not branching points. Moreover, the tangent cone at every branching point x is composed of a finite (and bounded by a constant depending on α and d) number of segments whose directions and masses satisfy (4.31). More precisely, ω satisfies in the sense of distributions on $[0, t_0]$ the elliptic equation*

$$-(|\omega|_{t, \eta}^\alpha \omega'(t))' = \sum_{i \in I} \theta_i^\alpha \delta_{\omega(t_i)} \hat{s}_i \quad (4.34)$$

where $\hat{s}_i$ is the tangent of the branch stemming from ω at the point $\omega(t_i)$ with mass θ_i . The fact that this tangent vector exists, and that the right hand side is a vector measure with finite mass, are guaranteed by the first parts of the statement.

Counterexamples to the existence of tangent directions can be built when the assumption on ν fails (see, again, [241]). For an alternative approach to the existence of tangent directions under slightly different assumptions, one can also see the last chapter in [271].

Numerics From the point of view of the practical search of the solutions, branched transport problems lie in between combinatorial optimization (they are similar to the

²¹Other Hölder results exist under different assumptions, if ν admits different dimensional lower bounds, i.e. it is Ahlfors regular of another exponent $k < d$, thus obtaining $\beta = d(\alpha - (1 - \frac{1}{k}))$, see [79].

Steiner minimal connection problem, which is, on a graph, known to be NP-hard) and continuous minimal flow problems. The numerical methods for finding a solution have to face at least two difficulties: concavity (and hence the high number of local minimizers) and singularity (i.e. working with singular structures such as networks instead of using L^p or smoother functions, as in the congested traffic case).

Some numerical results, based on local and non-local improvements of a suitably chosen initial guess, are present in [297] and [299]. Yet, here we prefer to introduce a completely different idea, based on a Γ -convergence approach.

Box 4.6. – Important notion – Γ -convergence

Γ -convergence is a theory, introduced by De Giorgi (see [139] and [134]), to provide approximations of minimization problems. It is also known as *epi-convergence*.

Definition - On a metric space X let $F_n : X \rightarrow \mathbb{R} \cup \{+\infty\}$ be a sequence of functions. We define the two lower-semi-continuous functions F^- and F^+ (called Γ -liminf and Γ -limsup of this sequence, respectively) by

$$F^-(x) := \inf_{n \rightarrow \infty} \{\liminf F_n(x_n) : x_n \rightarrow x\},$$

$$F^+(x) := \inf_{n \rightarrow \infty} \{\limsup F_n(x_n) : x_n \rightarrow x\}.$$

Should F^- and F^+ coincide, then we say that F_n actually Γ -converges to the common value $F = F^- = F^+$. The definition of Γ -convergence for a continuous parameter $\varepsilon \rightarrow 0$ obviously passes through the convergence to the same limit for any subsequence $\varepsilon_n \rightarrow 0$.

Among the properties of Γ -convergence we have the following:

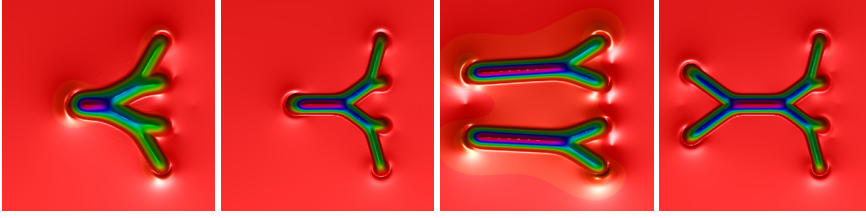
- if there exists a compact set $K \subset X$ such that $\inf_X F_n = \inf_K F_n$ for any n , then F attains its minimum and $\inf F_n \rightarrow \min F$;
- if $(x_n)_n$ is a sequence of minimizers for F_n admitting a subsequence converging to x , then x minimizes F ;
- if F_n is a sequence Γ -converging to F , then $F_n + G$ will Γ -converge to $F + G$ for any continuous function $G : X \rightarrow \mathbb{R} \cup \{+\infty\}$.

One of the first applications of Γ -convergence was to provide elliptic approximations of free discontinuity problems, or, more generally, variational problems with “singular energies”. We will only mention the following standard example (known as “Modica-Mortola” approximation, see [237] and [134]) because of its simplicity. Define the functional F_ε on $L^1(\Omega)$ through

$$F_\varepsilon(u) = \begin{cases} \frac{1}{\varepsilon} \int W(u(x)) dx + \varepsilon \int |\nabla u(x)|^2 dx & \text{if } u \in H^1(\Omega); \\ +\infty & \text{otherwise.} \end{cases}$$

Then, if $W(0) = W(1) = 0$ and $W(t) > 0$ for any $t \neq 0, 1$, we have Γ -convergence of the functionals F_ε towards the functional F given by

$$F(u) = \begin{cases} c\text{Per}(S) & \text{if } u = \mathbb{1}_S \text{ and } S \text{ is a finite-perimeter set;} \\ +\infty & \text{otherwise,} \end{cases}$$



where the constant c is given by $c = 2 \int_0^1 \sqrt{W(t)} dt$.

Inspired by this result, in [247] a similar result for branched transport has been proven. Unfortunately, it has only been proved for the two dimensional case $\Omega \subset \mathbb{R}^2$. Consider, on the space $\mathcal{M}_{\text{div}}^d(\Omega)$ of vector measures on Ω with divergence which is also a measure (endowed with the weak convergence of both the measures and their divergences), the functionals

$$M_\varepsilon^\alpha(\mathbf{w}) = \varepsilon^{\alpha-1} \int_\Omega |\mathbf{w}(x)|^\beta dx + \varepsilon^{\alpha+1} \int_\Omega |\nabla \mathbf{w}(x)|^2 dx, \quad (4.35)$$

defined on $\mathbf{w} \in H^1(\Omega; \mathbb{R}^2)$ and set to $+\infty$ outside $H^1 \cap \mathcal{M}_{\text{div}}^d(\Omega)$. The exponent $\beta = \frac{4\alpha-2}{\alpha+1}$ is fixed from suitable scaling computations.

Compared to the Modica-Mortola case, here the double-well potential is replaced with a concave power. Note that concave powers, in their minimization, if the average value for u is fixed in a region (which is in some sense the meaning of weak convergence), prefer either $u = 0$ or $|u|$ being as large as possible, i.e. there is sort of a double well at zero and infinity.

Theorem 4.46. *Suppose $d = 2$ and $\alpha \in]\frac{1}{2}, 1[$: then we have Γ -convergence of the functionals M_ε^α to cM^α , with respect to the convergence in $\mathcal{M}_{\text{div}}^d(\Omega)$, as $\varepsilon \rightarrow 0$, where c is a finite and positive constant (the value of c is actually $c = \alpha^{-1} (4c_0\alpha/(1-\alpha))^{1-\alpha}$, with $c_0 = \int_0^1 \sqrt{t^\beta - t} dt$).*

As a consequence of this approximation result, E. Oudet performed efficient numerical simulations by minimizing the functionals M_ε^α (more precisely: first a large value of ε is fixed, so that the problem is more or less convex, and a minimizer is found by gradient descent, then, following a continuation method, the value of ε is reduced and at every step the gradient descent starts from the previously found minimizer). In the end we find a “well-chosen” local minimizer, quite satisfactory in practice.

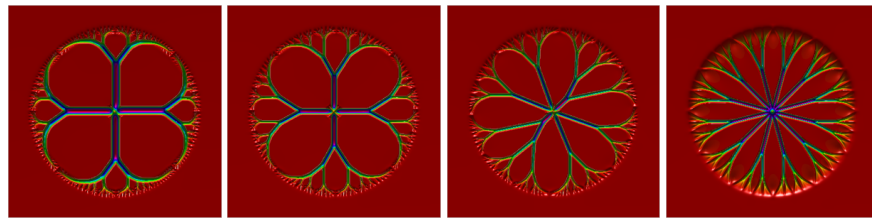


Figure 4.7: Branched transport computed via Γ -convergence between one point and four points, two points and four points, and one point and the uniform distribution on a circle, for different values of α (increasing from left to right). Pictures partially taken from [247], with permission.

Chapter 5

Wasserstein distances and curves in the Wasserstein spaces

In this chapter we will use the minimal value of transport problems between two probabilities in order to define a distance on the space of probabilities. We mainly consider costs of the form $c(x, y) = |x - y|^p$ in $\Omega \subset \mathbb{R}^d$, but the analysis can be adapted to a power of the distance in a more general metric space X . The exponent p will always be taken in $[1, +\infty[$ (we will briefly address the case $p = +\infty$ in the Discussion Section 5.5.1) in order to take advantage of the properties of the L^p norms. When Ω is unbounded we need to restrict our analysis to the following set of probabilities

$$\mathcal{P}_p(\Omega) := \left\{ \mu \in \mathcal{P}(\Omega) : \int_{\Omega} |x|^p d\mu < +\infty \right\}.$$

In a metric space, we fix an arbitrary point $x_0 \in X$, and set $\mathcal{P}_p(X) := \{ \mu \in \mathcal{P}(X) : \int d(x, x_0)^p d\mu(x) < +\infty \}$ (the finiteness of this integral does not depend on the choice of x_0).

Without entering into details for the moment, let us only mention that the distances that we use are defined in the following way: for $\mu, \nu \in \mathcal{P}_p(X)$, we set $W_p(\mu, \nu) := \mathcal{T}_c(\mu, \nu)^{1/p}$, where $\mathcal{T}_c$ is the minimal transport cost for $c(x, y) = d(x, y)^p$ and d is the distance on X .

The distance that we obtain in this way are called Wasserstein distances¹. They are very important in many fields of applications and they seem a natural way to describe distances between equal amounts of mass distributed on a same space. To translate into a language more meaningful for applications², one could say that Wasserstein distances are a way to take a distance onto a set X of objects (colors, words, points...) and create out of it a distance on the set of all possible histograms on X (see also Section 2.5.1).

¹The name is highly debated, in particular in Russia, since L. Vaserstein (whose name is sometimes spelled Wasserstein), a Russian scholar based in the United States, did not really play the key role that one could imagine in the introduction of these distances. Yet, this is nowadays the standard name in Western countries, probably due to the terminology used in [199, 246], and it seems impossible to change this convention, even if other names have been often suggested, such as Monge-Kantorovich distances...

²In particular in image processing. Thanks to G. Peyré for pointing out this interpretation.

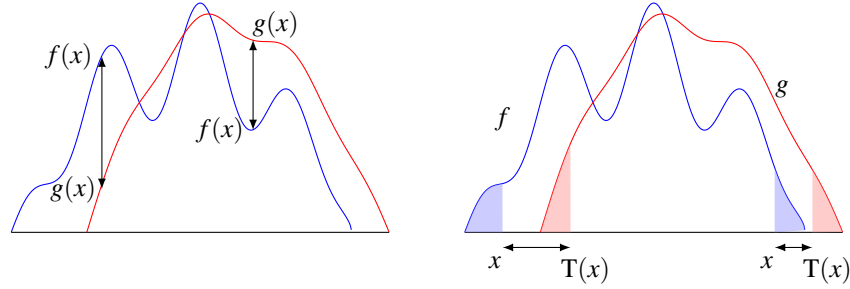


Figure 5.1: “Vertical” vs “horizontal” distances (the transport T is calculated in the picture on the right as in the 1D case, by imposing equality between the blue and red areas under the graphs of f and g , i.e. using the CFD functions as in Chapter 2).

It is interesting to compare these distances to L^p distances between densities (a comparison which is meaningful when we consider absolutely measures on $\mathbb{R}^d$, for instance). A first observation is the very different behavior of these two classes of distances. We could say that, if L^p distances can be considered “vertical”, Wasserstein distances are instead “horizontal”. This consideration is very informal, but is quite natural if one associates to every absolutely continuous measure the graph of its density. To compute $\|f - g\|_{L^p}$ we need to look, for every point x , the distance between $f(x)$ and $g(x)$, which corresponds to a vertical displacement between the two graphs, and then integrate this quantity. On the contrary, to compute $W_p(f, g)$ we need to consider the distance between a point x and a point $T(x)$ (i.e. an horizontal displacement on the graph) and then to integrate this, for a particular pairing between x and $T(x)$ which makes a coupling between f and g .

A first example where we can see the very different behavior of these two ways of computing distances is the following: take two densities f and g supported on $[0, 1]$, and define g_h as $g_h(x) = g(x - h)$. As soon as $|h| > 1$, the L^p distance between f and g_h equals $(\|f\|_{L^p}^p + \|g\|_{L^p}^p)^{1/p}$, and does not depend on the “spatial” information consisting in $|h|$. On the contrary, the W_p distance between f and g_h is of the order of $|h|$ (for $h \rightarrow \infty$) and depends much more on the displacement than on the shapes of f and g . As another example, consider the distance between a density g and its translation g_h , for $|h| \rightarrow 0$. In this case we have $W_p(g, g_h) = |h|$, while $\|g - g_h\|_{L^p}$ can be much larger than $|h|$, and is of order $|h|$ if and only if g is smooth enough.

In several applications, this robustness of the Wasserstein distances w.r.t. perturbation of the densities make them preferable to other distances, and the fact that they quantify spatial features of the measures is exactly what is required. Also, we will see (in Section 5.4 and later 6.1 and 7.3), that the geodesics for these distances provide useful interpolations between distributions of mass.

5.1 Definition and triangle inequality

In this section we give the basic definitions of Wasserstein distances and spaces in $\Omega \subset \mathbb{R}^d$. We will see how to adapt the analysis to the case of more general metric spaces. As we said above, we restrict our analysis to the set

$$\mathcal{P}_p(\Omega) := \left\{ \mu \in \mathcal{P}(\Omega) : \int |x|^p d\mu < +\infty \right\}.$$

Note that $p < q$ implies $\mathcal{P}_p(\Omega) \subset \mathcal{P}_q(\Omega)$ and that, whenever Ω is bounded, $\mathcal{P}_p(\Omega) = \mathcal{P}(\Omega)$.

For $\mu, \nu \in \mathcal{P}_p(\Omega)$, let us define

$$W_p(\mu, \nu) := \min \left\{ \left(\int_{\Omega \times \Omega} |x - y|^p d\gamma : \gamma \in \Pi(\mu, \nu) \right)^{\frac{1}{p}}, \right.$$

i.e. the p -th root of the minimal transport cost $\mathcal{T}_c$ for the cost $c(x, y) = |x - y|^p$. The assumption $\mu, \nu \in \mathcal{P}_p(\Omega)$ guarantees finiteness of this value since $|x - y|^p \leq C(|x|^p + |y|^p)$, whence $W_p^p(\mu, \nu) \leq C(\int |x|^p d\mu + \int |x|^p d\nu)$.

Note that, due to Jensen inequality, since all $\gamma \in \Pi(\mu, \nu)$ are probability measures, for $p \leq q$ we have

$$\left(\int |x - y|^p d\gamma \right)^{\frac{1}{p}} = \|x - y\|_{L^p(\gamma)} \leq \|x - y\|_{L^q(\gamma)} = \left(\int |x - y|^q d\gamma \right)^{1/q},$$

which implies $W_p(\mu, \nu) \leq W_q(\mu, \nu)$. In particular $W_1(\mu, \nu) \leq W_p(\mu, \nu)$ for every $p \geq 1$. We will not define here W_∞ , but we refer to Section 5.5.1 for a definition as a limit for $p \rightarrow \infty$.

On the other hand, for bounded Ω an opposite inequality also holds, since

$$\left(\int |x - y|^p d\gamma \right)^{\frac{1}{p}} \leq \text{diam}(\Omega)^{\frac{p-1}{p}} \left(\int |x - y| d\gamma \right)^{\frac{1}{p}}.$$

This implies

$$W_p(\mu, \nu) \leq C W_1(\mu, \nu)^{\frac{1}{p}}, \quad (5.1)$$

for $C = \text{diam}(\Omega)^{(p-1)/p}$.

Proposition 5.1. *The quantity W_p defined above is a distance over $\mathcal{P}_p(\Omega)$.*

Proof. First, let us note that $W_p \geq 0$. Then, we also remark that $W_p(\mu, \nu) = 0$ implies that there exists $\gamma \in \Pi(\mu, \nu)$ such that $\int |x - y|^p d\gamma = 0$. Such a $\gamma \in \Pi(\mu, \nu)$ is concentrated on $\{x = y\}$. This implies $\mu = \nu$ since, for any test function ϕ we have

$$\int \phi d\mu = \int \phi(x) d\gamma = \int \phi(y) d\gamma = \int \phi d\nu.$$

We need now to prove the triangle inequality. We give two different proofs of this fact. The first (Lemma 5.3) is easier and uses transport maps and approximation; the second (Lemma 5.4) is more general but requires a trickier tool, i.e. disintegrations of measures. □ □